

ABHIJNAN NATH

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EDUCATION

PhD in Computer Science |

Jan '23 – May '26

Colorado State University, Fort Collins, CO

Fort Collins, CO

Dissertation: The Generative Friction Trilogy: Learning to Intervene, Evaluate, and Collaborate in Multi-Agent Collaboration

Conferred: May 16, 2026 | Defended: February 26, 2026 | Advisor: Dr. Nikhil Krishnaswamy

MS in Computer Science | AI/NLP

Jan '21 – Dec '22

Colorado State University, Fort Collins, CO

Fort Collins, CO

Advisor: Dr. Nikhil Krishnaswamy

M.Sc. (Integrated with BS) | Physics

Aug '09 – Aug '13

Birla Institute of Technology and Science, Pilani

Pilani, India

WORK EXPERIENCE

Applied ML Research Scientist

May '26 – Present

Brand Intelligence, Adobe Inc.

Seattle, WA

- Designed continual-learning data pipelines that incorporate end-user feedback for brand compliance validation in multi-agent, long-horizon learning setups.

Graduate Research Assistant

Dec '21 – May '26

SIGNAL Lab, Colorado State University

- Supported by DARPA FACT and AIDA-RAMFIS programs, NSF DRL/IIS, U.S. Army Research Office (ARO), and ARPA-H PARADIGM programs.*
- Led epistemic decision process (EDP) framework and ECHO, a turn-level policy gradient method with belief-sensitive credit assignment for multi-turn LLM agents; trained a 1.5B model that outperforms frontier models on sequential information-seeking tasks.
- Led CRAFT, a grounded multi-agent coordination benchmark evaluating pragmatic communication under partial information revealing failure modes in frontier LLMs.
- Led FAAF (Friction-Aware Alignment Framework) for training LLMs to generate productive interventions in collaborative settings; highlighted in [Microsoft New Future of Work Report 2025](#) as a key direction for collaborative AI systems.
- Deployed FAAF-based real-time friction intervention system for the NSF iSAT program, providing live AI-driven collaborative learning support in middle school STEM classrooms.
- Led ICR (Interruptible Collaborative Roleplayer), formalizing partner-aware learning for multi-agent LLM systems; addresses systematic failure of standard RLHF-trained agents to respond to well-meaning interventions.
- Led LLM alignment pipelines including simultaneous reward distillation and diverse preference learning.
- Led AxomiyaBERTa, the first monolingual phonologically-aware transformer model for Assamese, a low-resource language
- Led coreference resolution systems for events and entities using linear semantic transfer and mixed-modality ensembles

Applied Scientist II Intern

July '25 – Oct '25

Amazon Science

- Designed novel **causal credit assignment** algorithms for LLM-based search and recommendation, improving reward attribution and retrieval quality.
- Led **multi-GPU scaling and optimization** of RL training pipelines, achieving superior performance over state-of-the-art baselines such as GRPO across multiple product recommendation benchmarks.

PhD Research Intern

April '25 – July '25

Cresta Intelligence Inc.

- Email Agent with Reinforcement Learning: Implemented an agentic framework for a RAG-enhanced email writing system for Square.
- Led experimental work for upgrading base foundational LLMs for customer calls summarization.

PhD Research Intern

May '24 – Oct '24

Optum AI/UnitedHealth Group

- LLM Post-training: Preference Alignment in LLMs with Supervised Learning
- Developed novel alignment algorithms for Diverse Preference Learning in healthcare

LLM Research Intern

May '23 – Aug '23

Sprouts AI

- Designed, optimized, and productionized various LLM pipelines for personalized outreach
- Productionized in-context learning frameworks using sequencing platforms like Langchain and Knowledge-Retrieval systems with RAG.

HONORS AND AWARDS

Recognized as Silver Reviewer, ICML 2026

International Conference on Machine Learning (ICML), Seoul, South Korea

Awarded Wim Böhm Ph.D. Fellowship in Computer Science, 2025

Department of Computer Science, Colorado State University, Fort Collins, CO

Evolutionary Computing and Artificial Intelligence Graduate Fellowship 2024

Department of Computer Science, Colorado State University, Fort Collins, CO

Best Student Paper Award

International Conference on Educational Data Mining (EDM) 2024, Atlanta, GA

PhD Candidacy with Distinction, December 2024

Colorado State University, Fort Collins, CO

PUBLICATIONS AND PREPRINTS

1. **ECHO: Learning Epistemically Adaptive Language Agents with Turn-Level Credit**
Abhijnan Nath, Nikhil Krishnaswamy
[preprint](#)
2. **CRAFT: Grounded Multi-Agent Coordination Under Partial Information**
Abhijnan Nath, Hannah VanderHoeven, Nikhil Krishnaswamy
[preprint](#)
3. **Owen-Shapley Policy Optimization: A Principled RL Algorithm for Generative Search LLMs**
Abhijnan Nath, Alireza Bagheri Garakani, Tianchen Zhou, Fan Yang, Yan Gao, Nikhil Krishnaswamy
[preprint](#)
4. **Collaborate, Deliberate, Evaluate: How LLM Alignment Affects Coordinated Multi-Agent Outcomes**
Abhijnan Nath, Carine Graff, Nikhil Krishnaswamy
(Accepted at AAMAS 2026). [paper](#)

5. **Learning “Partner-aware” Collaborators in Multi-Party Collaboration**
Abhijnan Nath, Nikhil Krishnaswamy
(NeurIPS 2025). [paper](#)
6. **Frictional Agent Alignment Framework: Slow Down and Don’t Break Things**
Abhijnan Nath, Carine Graff, Andrei Bachinin, Nikhil Krishnaswamy
(ACL Main 2025). [paper](#)
7. **Dynamic Epistemic Friction in Dialogue**
Timothy Obiso, Kenneth Lai, Abhijnan Nath, Nikhil Krishnaswamy, James Pustejovsky
(ACL 2025 Workshop CoNLL). [paper](#)
8. **Simultaneous Reward Distillation and Preference Learning: Get You a Language Model Who Can Do Both**
Abhijnan Nath, Changsoo Jung, Ethan Seefried, Nikhil Krishnaswamy
[preprint](#)
9. **DPL: Diverse Preference Learning Without A Reference Model**
Abhijnan Nath, Andrey Volozin, Saumajit Saha, Albert Aristotle Nanda, Galina Grunin, Rahul Bhotika, Nikhil Krishnaswamy
(Oral, Accepted at NAACL Main 2025). [paper](#)
10. **“Any Other Thoughts, Hedgehog?” Linking Deliberation Chains in Collaborative Dialogues**
Abhijnan Nath, Videep Venkatesha, Mariah Bradford, Avyakta Chelle, Austin Collin Youngren, Carlos Mabrey, Nathaniel Blanchard, Nikhil Krishnaswamy.
(Published at Findings of EMNLP 2024). [paper](#)
11. **Okay, Let’s Do This! Modeling Event Coreference with Generated Rationales and Knowledge Distillation**
Abhijnan Nath, Shadi Manafi, Avyakta Chelle, and Nikhil Krishnaswamy.
(Oral, Published at NAACL Main 2024). [paper](#)
12. **Multimodal Cross-Document Event Coreference Resolution Using Linear Semantic Transfer and Mixed-Modality Ensembles**
Abhijnan Nath, Huma Jamil, Shafiuddin Rehan Ahmed, George Baker, Rahul Ghosh, James H. Martin, Nathaniel Blanchard, and Nikhil Krishnaswamy.
(Published at LREC-COLING 2024). [paper](#)
13. **Propositional Extraction from Natural Speech in Small Group Collaborative Tasks**
Videep Venkatesha, Abhijnan Nath, Ibrahim Khebour, Avyakta Chelle, Mariah Bradford, Jingxuan Tu, James Pustejovsky, Nathaniel Blanchard, and Nikhil Krishnaswamy
(Best Paper Award at EDM 2024).
14. **AxomiyaBERTa: A Phonologically-aware Transformer Model for Assamese**
Abhijnan Nath, Sheikh Mannan, and Nikhil Krishnaswamy
(Published at Findings of ACL 2023). [paper](#)

15. **2^n is better than n^2 : Decomposing Event Coreference Resolution into Two Tractable Problems**
Shafiuddin Rehan Ahmed, Abhijnan Nath, James H. Martin, and Nikhil Krishnaswamy
(Published at Findings of ACL 2023). [paper](#)
16. **How Good is the Model in Model-in-the-loop Event Coreference Resolution Annotation?**
Shafiuddin Rehan Ahmed, Abhijnan Nath, Michael Regan, Adam Pollins, Nikhil Krishnaswamy, and James H. Martin
(Published at the Linguistic Annotation Workshop, ACL 2023). [paper](#)
17. **A Generalized Method for Automated Multilingual Loanword Detection**
Abhijnan Nath, Sina Mahdipour Saravani, Ibrahim Khebour, Sheikh Mannan, Zihui Li, and Nikhil Krishnaswamy
(Published at COLING 2022) [paper](#)
18. **Phonetic, Semantic, and Articulatory Features in Assamese-Bengali Cognate Detection**
Abhijnan Nath, Rahul Ghosh, and Nikhil Krishnaswamy
(Published at VarDial 2022) [paper](#)
19. **Linear Mappings: Semantic Transfer from Transformer Models for Cognate Detection and Coreference Resolution**
Abhijnan Nath
MS Thesis. Colorado State University, 2022. [paper](#)

TEACHING EXPERIENCE

Graduate Teaching Assistant CS 445: Introduction to Machine Learning	Spring 2024 Colorado State University
Graduate Teaching Assistant CS 542: Natural Language Processing	Fall 2023 Colorado State University
Graduate Teaching Assistant CS 320: Algorithm Theory and Practice	Fall 2021 Colorado State University